

The design includes an AI-enabled processor capable of tracking and destroying multiple threats simultaneously. The framework is organised around three closely linked layers: the sensor layer, the command-and-control layer, and the interceptor layer.

Together, these layers sustain a continuous cycle of surveillance, AI-enabled decision-making processes, and engagement, enabling rapid response and operational effectiveness. Supporting the layers are auxiliary systems, including electronic warfare units, cyber defence mechanisms, satellite-based communication systems, and secure logistics and power infrastructure.

Supported by auxiliary systems, the structure enables seamless integration of future capabilities without compromising operational readiness.

Sensor layer

The sensor layer forms the surveillance backbone, integrating space-based, airborne, and ground-based sensors to provide continuous radar and data coverage across national airspace.

Operating in sync, long-range active electronically scanned array (AESA) radars (Box 1), airborne early-warning platforms, and surveillance satellites provide continuous target detection, early warning, and situational awareness

even during electronic or cyber disruption.

Command-and-control layer

The command-and-control layer, anchored by the integrated air command and control system (IACCS), functions as the central nervous system. The layer fuses continuous streams of data from civil and military radars, satellites, and electronic intelligence sources. The fused raw data is transformed into a unified operational picture and instantly shared across the Army, Navy, and Air Force.

The AI system evaluates threat profiles by correlating sensor tracks, assessing intent, and prioritising risks. The validated threats are translated into engagement directives for the interceptor layer. Secure, encrypted communication networks protected against jamming and electronic warfare ensure uninterrupted command and control in hostile environments.

Interceptor layer

The interceptor layer is responsible for neutralising hostile targets using assets such as the integrated rocket force (IRF), precision-guided munitions, hypersonic missiles, and directed-energy weapons.

Once a threat is identified, the interceptor system (Box 2) engages the threat and provides a protective shield for the sensor and command-and-control layers, closing the operational loop. Operational feedback, including success rates, resource utilisation, and enemy response, is relayed to the command-and-control layer for real-time reassessment, rapid resource reallocation, and simultaneous counter-threat operations.

Mission Sudarshan Chakra: Implementation

The Ministry of Defence is designated as the lead agency for implementing Mission Sudarshan Chakra, with the IAF as the nodal service and the DRDO as the principal technical

Box 1: Active electronically scanned array (AESA) radars

AESA is an advanced radar technology built around thousands of miniature solid-state modules. AESA radar electronically steers beams, enabling simultaneous multi-target tracking and rapid transitions between surveillance and engagement.



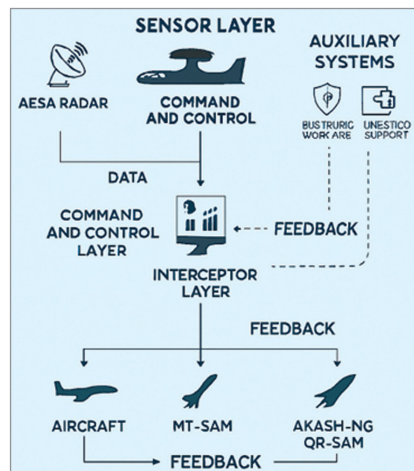
An AESA radar system (Source: <https://www.aviationtoday.com>)

agency. This ensures that operational leadership, technological innovation, and strategic oversight function together within a unified framework.

The mission is executed through a building-block approach, allowing capabilities to be developed, tested, and integrated step by step. The implementation is divided into two phases. The first phase focuses on the core architecture, sensor integration, command-and-control networks, and baseline interceptor capabilities. The second phase expands and deepens the system through advanced technologies. This phased approach ensures that lessons learned are reinvested to reduce risks and build a resilient, adaptive national defence shield.

Phase I: Foundational infrastructure and system integration

(2025-2030). Phase I focuses on establishing foundational infrastructure and integrating core systems. This includes setting up command-and-control centres, building secure communication networks, and linking radar, sensors, and interceptor



Mission Sudarshan architecture (Source: ChatGPT)